

Automotive Network Monitor System

Test cases and Results

#	Type	Test Scenario	Test Steps	Expected Result	Result
1	Functional	Server starts and offers a transport choice	Launch the server executable.	Server console shows a menu asking to choose TCP or UDP, and confirms the selected mode.	Passed
2	Functional	A TCP client connects to the server	Start the server in TCP mode, then start one client in TCP mode.	Client console shows it connected successfully; server console shows a client connected and the total connected count.	Passed
3	Functional	A UDP client registers with the server	Start the server in UDP mode, then start one client in UDP mode.	Client console shows it registered successfully; server console shows the client name and total connected count.	Passed
4	Functional	Multiple clients connect at the same time	Start the server, then start all three clients (Gateway, Diagnostic, DataLogger) one after another.	Server console shows the connected count increasing to 3, with each client's name listed as it joins.	Passed
5	Positive	Server sends events on its own, without needing a new client	With all clients already connected, wait without starting any new client or pressing any key.	After a short wait, every connected client console shows a new event arriving on its own, with no new client having connected.	Passed
6	Positive	Client shows event details clearly	Observe a client console when any event is received.	The client console clearly displays the event ID, event type (e.g. "Engine Start"), severity (e.g. "HIGH"), and which ECU sent it.	Passed
7	Positive	Client acknowledges every event it receives	Observe both the client and server console after any event is delivered.	The client console confirms it sent an acknowledgement; the server console confirms it received that acknowledgement, showing the client name and a success status.	Passed
8	Negative	A repeated event is recognized and not processed twice	Trigger the same event a second time (using the server's repeat/duplicate command).	The client console clearly states the event was already received and is being ignored, while the server console still shows an acknowledgement was sent back.	Passed

#	Type	Test Scenario	Test Steps	Expected Result	Result
9	Negative	System keeps running if a client disconnects unexpectedly	With a client connected, close that client's window/process directly.	The server console shows a failure message for that specific client, but continues normally and keeps working with the remaining clients.	Passed
10	Negative	Server enforces a maximum number of clients	Connect the maximum supported number of clients (10), then attempt to connect one more.	The server console shows a message that the client limit has been reached and the extra connection is rejected, without the server crashing.	Passed

Legend: Functional = directly verifies a stated requirement. Positive = expected behavior under normal conditions. Negative = expected behavior under duplicate, boundary, or failure conditions. All cases above were executed manually against the built executables in both TCP and UDP modes prior to submission.